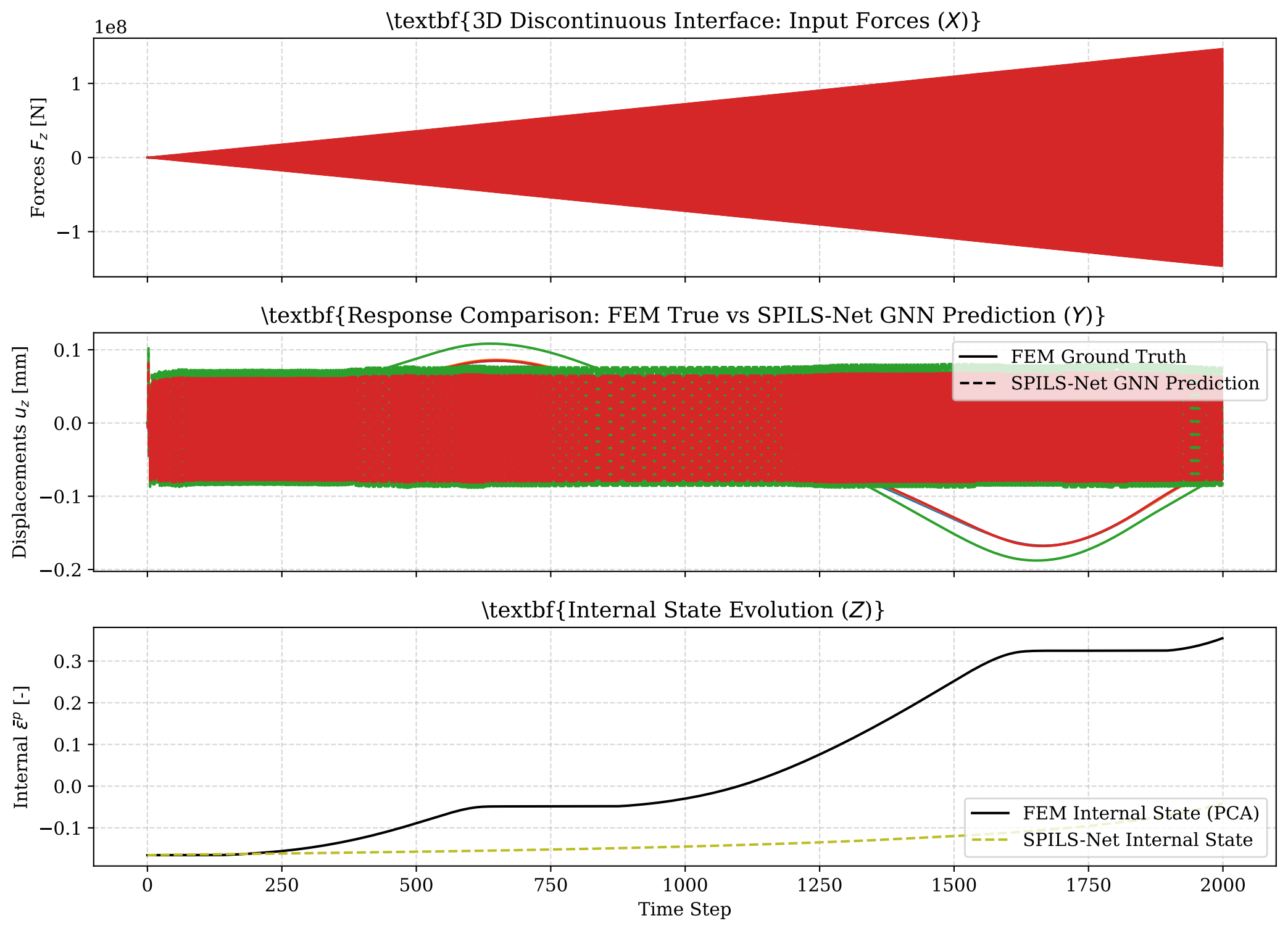

\textbf{3D Discontinuous Interface: Input Forces (X)}
1e8
Forces F_z [N]
1
0
−1
\textbf{Response Comparison: FEM True vs SPILS-Net GNN Prediction (Y)}
Displacements u_z [mm]
0.1
0.0
−0.1
−0.2
FEM Ground Truth
SPILS-Net GNN Prediction
\textbf{Internal State Evolution (Z)}
Internal $\bar{\varepsilon}^p$ [-]
0.3
0.2
0.1
0.0
−0.1
FEM Internal State (PCA)
SPILS-Net Internal State
0
250
500
750
1000
1250
1500
1750
2000
Time Step